

Note: All problems are worth 100 points. Your final score will be the average value assignment #6

Course: CHE 341. CHEMICAL PROCESS CONTROL

Term: Spring 2020

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Due Date: Saturday April 11th by 12:00 noon

Number of points.....100

Your score.....

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

Scan your Assignment and create a PDF file

PDF File Name: Assignment6_XY.PDF o X: First letter of your first name

o Y: Your last name

o Example: Assignment6_MCaracotsios.PDF

Email your PDF file to: Daniel Christiansen dchris28@uic.edu and Yuechen Gao ygao54@uic.edu by the due date (Saturday April 11) and time (12:00 noon).

Problem 6A:

The goal of this assignment is to arrange a feedback control loop that controls the temperature of the reactor unit. A disturbance is from the cooling water jacket that affects the reactor temperature.

Below are tables that show the method, parameters, PI-Control and the PID

Table 1: Z-N method cooling water

Method	Parameter	PI- control	PID
Z-N	KC	-3.95	-5.27
	TI	17.11	10.27
	TD	-	2.56

Table 2: Cohen-Coon method cooling water

Method	Parameter	PI- control	PID
Cohen-Coon	KC	-0.301	-6.39
	TI	9.42	11.02
	TD	-	1.76

Table 3: Lambda Tuning cooling water

Method	Parameter	PI- control	PID
Lambda Tuning	KC	-3.69	-2.6
	TI	12.24	12.24
	TD	-	2.57
	Lambda	1	1

Table 4: Cycling tuning cooling water

Method	Parameter	PI- control	PID
Cycling Tuning	KC	-17.79	-23.73
	TI	15.199	9.16
	TD	-	2.28
	KU	-79.1	-
	PU	18.31	-

Table 5: Z-N feed temperature

Method	Parameter	PI- control	PID
Z-N	KC	14.17	18.89
	TI	14.75	8.86
	TD	-	2.21

Table 6: Cohen-Coon feed temperature

Method	Parameter	PI- control	PID
Cohen-Coon	KC	0.20	22.31
	TI	9.61	9.98
	TD	-	1.56

Table 7: Lambda tuning feed temperature

Method	Parameter	PI- control	PID
Lambda Tuning	KC	8.468	6.673
	TI	16.46	16.46
	TD	-	2.21
	Lambda	1	1

Table 7: cycling tuning

Method	Parameter	PI- control	PID
Cycling Tuning	KC	-17.79	-23.73
	TI	15.19	9.16
	TD	-	2.29
	KU	-79.11	-
	PU	18.31	-

The Z-N method is called quarter-amplitude damping the tuning rule calculate the gain and dose not calculate the proportional bond. The Cohen-Coon method works very well in most cases. The lag-dominate process they apply (td) equal or less than (4 tou). Lambda tuning rule not necessarily PID controller.

If one cannot measure the dead time accurately should use the lambda tuning rule, as for the process to respond a specific time constant one should use the lambda tuning rule.

Using the CHE341 calculator one would be able to plot controller 1, controller 2, controller 3, and feed-forward (PV vs time).

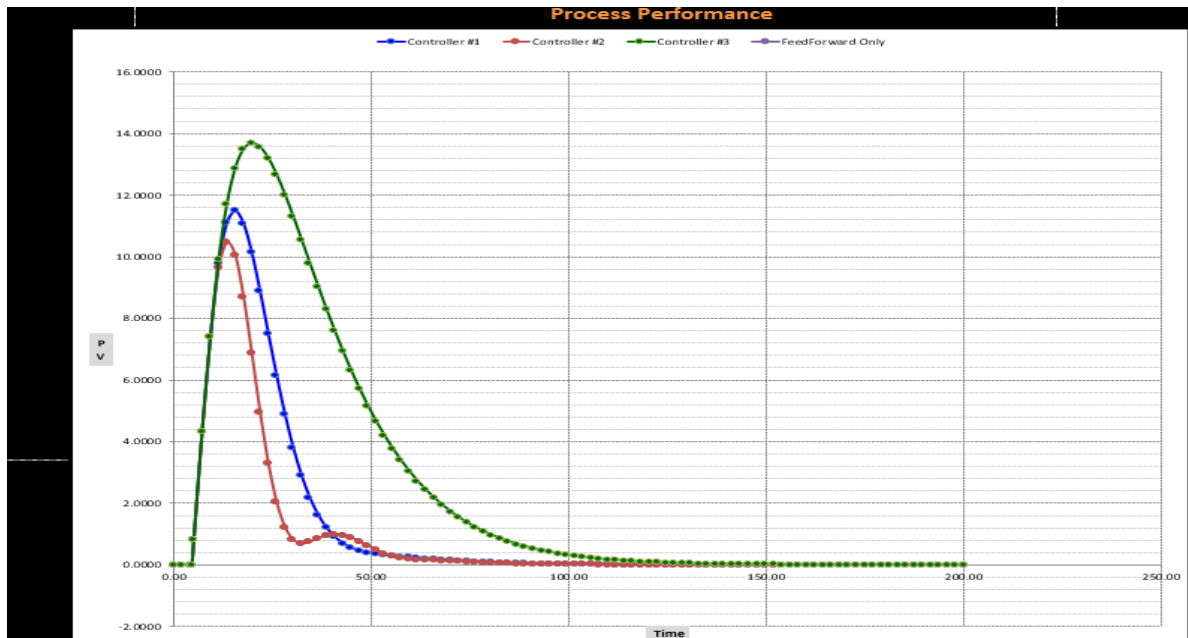


Figure 1: PID controller

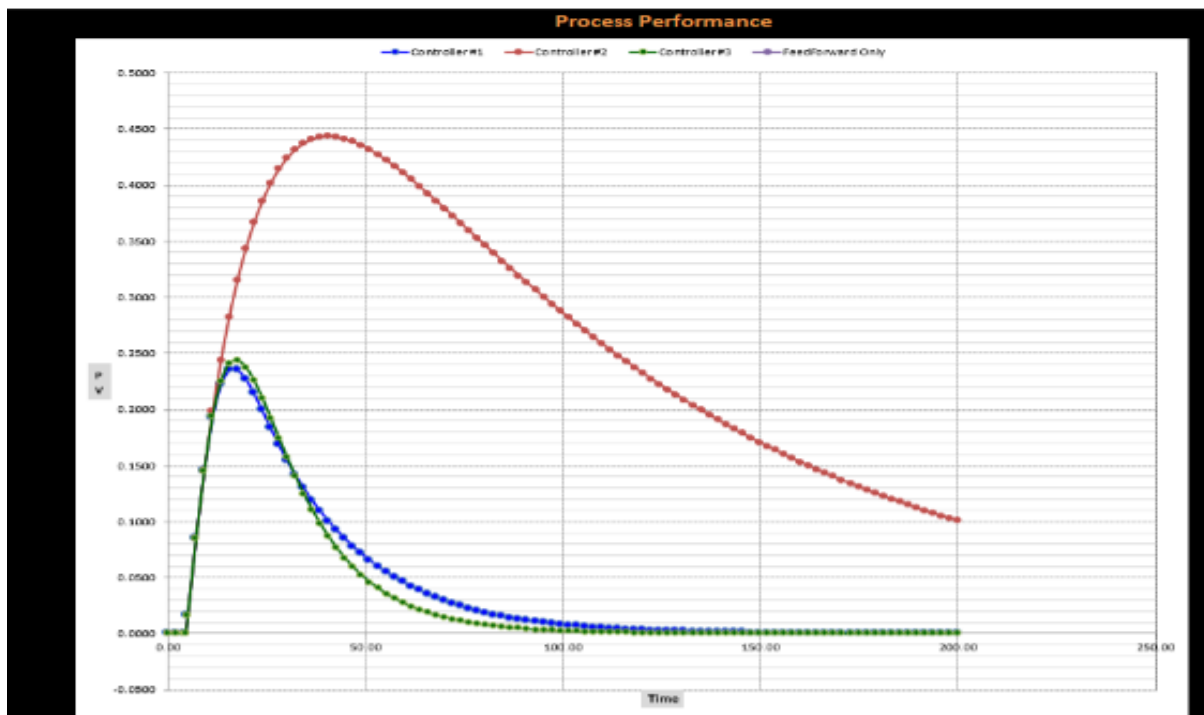


Figure 2: PI controller

Code used in this assignment:

```
!Variable Declarations and Data
!*****
Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Q, Tfeed, CAfeed, CEfeed, CEAfeed, CWfeed As Real
Q      = 454.2      !L/mins
Tfeed  = 298.15 !K
CAfeed = 4005.44 !mol/min
CEfeed = 3875.97 !mol/min
CEAfeed = 0 !mol/min
CWfeed = 210.39 !mol/min

Global V, Vj, Ua As Real
V = 10000 !L Reactor Volume
Vj = 2000 !L Jacket cooling volume water from residence time
Ua = 1012000 !J/(min K)

Global rho, rhoj, Cp, Cpj, DHr, Cv As Real
rho = 17.8 !mol/L
rhoj = 54.5 !mol/L
Cp = 116.76 !J/(mol K)
Cpj = 75 !J/(mol K)
DHr = -29070 !J/mol
Cv = 36 !Setpoint 60% Opening

Global K01, K02, Er As Real
K01 = 1.14E+10 !L/mol min
K02 = 3.00E+09 !L/mol min
Er = 7156.172 !K

@Initial Conditions
!U(1)= 0.5*rho
!U(2)= 0
!U(3)= 0
!U(4)= 0.5*rho
!U(5)= 298.15
!U(6)= 298.15
!U(7)= 298.15

!Steady State Condition
U(1) = 3.16575E+00
U(2) = 2.88070E+00
U(3) = 5.65292E+00
U(4) = 6.11613E+00
U(5) = 3.46353E+02
U(6) = 3.17563E+02
U(7) = 3.46353E+02

@Model Equations
Dim CA, CE, CEA, CW, Temp, TCj, K1, K2, R1, Qj, Tm As Real
CA = U(1)
CE = U(2)
CEA = U(3)
CW = U(4)
Temp = U(5)
TCj = U(6)
Tm = U(7)
!TCj = 316.15 !K
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```

K1 = K01*exp(-Er/Temp) !L/(mol*min)
K2 = K02*exp(-Er/Temp) !L/(mol*min)
R1 = K1*CA*CE-K2*CEA*CW !mol/(L*min)
Qj = 367.18 !L/min F
!Qj = Cv *10.2 !L/min

If(iRange==1) TCj= 293.15 !
!If(iRange==1) Tfeed = 303.15 !
!If(iRange==1) CWfeed = 400 !

F(1)=CAfeed-Q*CA+(-R1)*V
F(2)=CEfeed-Q*CE+(-R1)*V
F(3)=CEAfeed-Q*CEA+(R1)*V
F(4)=CWfeed-Q*CW+(R1)*V
F(5)=Q*rho*Cp*(Tfeed-Temp)+(-DHR)*R1*V-Ua*(Temp-TCj)
F(6)=Qj*rhoj*Cpj*(Tfeed-U(6))+Ua*(Temp-U(6))
F(7)= -U(7)+U(5)
@Coefficient Matrix
E(1)= V
E(2)= V
E(3)= V
E(4)= V
E(5)= rho*Cp*V
E(6)= rhoj*Cpj*Vj
E(7) = 5
@Solver Options
Neq = 7
Npts = 50
Tend = 150
Tstop = 1
Param(28)=1
VariableNames=Time [min]; CA [mol/L]; CE [mol/L]; CEA [mol/L]; CW [mol/L]; Reactor Temp [K]; TCj [K]

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